

Paridhi Rustogi

POSTDOCTORAL RESEARCH SCIENTIST · COLUMBIA UNIVERSITY

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Research Interests

I am a biogeochemical oceanographer studying the ocean's carbon and oxygen cycles. My research focuses on the influence of wave breaking and bubble-mediated air-sea gas exchange, with implications for quantifying the ocean carbon sink, understanding ocean deoxygenation, and evaluating emerging marine carbon dioxide removal strategies.

Education

Doctor of Philosophy in Earth Sciences (Incidental Master of Arts)

PRINCETON UNIVERSITY

2021–2026

United States

- Advisor: Prof. Laure Resplandy. Committee: Prof. Luc Deike, Prof. Curtis Deutsch, Prof. Daniel Sigman
- Thesis: Evaluating the impact of bubble-mediated air-sea gas exchange on carbon dioxide and oxygen in the ocean

Masters in Integrated Climate System Sciences

UNIVERSITY OF HAMBURG

2018–2021

Germany

- Advisor: Prof. Dr. Johanna Baehr, Dr. Peter Landschützer, Dr. Sebastian Brune
- Thesis: Seasonal analysis of atmosphere-ocean carbon flux variability in the North Atlantic and the Southern Ocean

Bachelor of Technology in Environmental Engineering

DELHI TECHNOLOGICAL UNIVERSITY

2013–2017

India

- Advisor: Prof. Anil Kumar Haritash
- Thesis: Comparative analysis of air quality in major cities of India

Publications

PEER-REVIEWED

6. **Rustogi, P.**, Resplandy, L., Bushinsky, S., and Deike, L. (2026). The influence of waves and bubbles on oxygen in the ocean interior. *Environmental Research Letters* 21 (9). doi:10.1088/1748-9326/ae612b [Special issue on Ocean Deoxygenation]
5. Deike, L., Zhou, X., **Rustogi, P.**, Stanley, R., Bushinsky, S., Reichl, B., and Resplandy, L. (2025). A universal wind-wave-bubble formulation for air-sea gas exchange and its impact on oxygen fluxes. *PNAS* 122 (38) e2419319122. doi:10.1073/pnas.2419319122. [News] [2025 Cozzarelli Prize Finalist]
4. **Rustogi, P.**, Resplandy, L., Liao, E., Reichl, B., and Deike, L. (2025). Influence of wave-induced variability on ocean carbon uptake. *AGU Global Biogeochemical Cycles* 39. doi:10.1029/2024GB008382 [Editor's Highlight Paper]
3. **Rustogi, P.**, Landschützer, P., Brune, S., and Baehr, J. (2023). The impact of seasonality on the annual air-sea carbon flux and its interannual variability. *npj Clim. Atmos. Sci.* 6(1). doi:10.1038/s41612-023-00378-3 [Blog] [News]
2. **Rustogi, P.** and Singh, S. K. (2017). Ecological benefits of reviving urban water bodies using rainwater harvesting. *Int. J. Adv. Res. Innov.* 5 (2). doi:10.51976/ijari.521713
1. **Rustogi, P.** and Singh, S. K. (2017). Revival and rejuvenation strategy of water bodies in a metropolitan city: a case study of Najafgarh lake, Delhi, India. *Int. J. Adv. Res.* 5 (2). doi:10.21474/IJAR01/3131

NON-AUTHOR CONTRIBUTIONS

Contributing analyst and model user/developer, acknowledged in the following Global Carbon Budget (GCB) reports:

- GCB 2025: Friedlingstein et al. (2025). *Earth Syst. Sci. Data*. doi:10.5194/essd-2025-659.
- GCB 2024: Friedlingstein et al. (2024). *Earth Syst. Sci. Data*. doi:10.5194/essd-17-965-2025.
- GCB 2024: Friedlingstein et al. (2023). *Earth Syst. Sci. Data*. doi:10.5194/essd-15-5301-2023.

Reviewer, Intergovernmental Panel on Climate Change Sixth Assessment Report (AR6) – Second draft:

- Working Group II – Chapter 3: Ocean and Coastal Ecosystems and their Services (2021)
- Working Group I – Chapter 2: Changing State of the Climate System (2020)

POLICY & APPLIED RESEARCH CONTRIBUTIONS

5. **Rustogi, P.** and Pervaiz, A. (2018). Integrated Water Resources Management Practices in Greater and Lesser Cholistan, Pakistan. GWP Knowledge Resources #497. [Link]
4. **Rustogi, P.**, Wangdi, K. and Lhendup, U. (2018). Siphon Irrigation for Farmer Communities in Lingmutey-Chu Basin in Bhutan. GWP Knowledge Resources #496. [Link]
3. **Rustogi, P.** and Athukorala, K. (2017). Disaster Risk Reduction through Climate Change Adaptation in Aranayake, Sri Lanka. GWP Knowledge Resources #495. [Link]
2. **Rustogi, P.** and Amir, P. (2017). IWRM Practices to Alleviate Poverty – A Model of Desert Development in Tharparkar, Pakistan. GWP Knowledge Resources #493. [Link]
1. **Rustogi, P.**, Khanduri, V. and Gurunathan, A. (2017). Augmenting Water Security and Food Security of Small Farmers in Andhra Pradesh, Karnataka and Tamil Nadu. GWP Knowledge Resources #491. [Link]

Professional Experience

Columbia University

USA

POSTDOCTORAL RESEARCH SCIENTIST, LAMONT-DOHERTY EARTH OBSERVATORY

2026-

- Investigating bubble-mediated air–sea CO₂ and O₂ exchange using ASTE-BGC, a high-resolution, data-assimilated North Atlantic biogeochemical ocean model.

Princeton University

USA

POSTDOCTORAL RESEARCHER, DEPARTMENT OF GEOSCIENCES

2026-

- Quantified the impact of physical transport on Thorium-234 data from the GEOTRACES GP17-ANT cruise in the Amundsen Sea.

University of Hamburg (UHH)

Germany

RESEARCH ASSISTANT, INSTITUTE OF OCEANOGRAPHY

2019 - 2021

- Applied statistical analyses of air–sea CO₂ fluxes using MPI's ESM (MPI-ESM) Grand Ensemble (MPI-GE) output and assimilation (EnKF), benchmarking against an observation-based pCO₂ flux product (SOM-FFN). [Interview]

International Water Management Institute (IWMI)

Sri Lanka

INTERN, RESOURCE RECOVERY AND REUSE GROUP

2018

- Curated and systematized a global database of educational resources on waste resource, recovery, and reuse, supporting capacity-building initiatives in higher education.

Global Water Partnership (GWP)

Sri Lanka

INTERN, SOUTH ASIA REGIONAL OFFICE

2017-2018

- Authored five case studies on climate adaptation and water security across South Asia, disseminated through GWP's international Water and Climate Development Program (WACDEP) initiative to inform regional policy and practice. [Blog post]

Indian Institute of Technology Delhi (IIT-D)

India

CONSULTANT, CENTRE OF EXCELLENCE IN CLIMATE MODELING

2017

- Contributed to manuscript preparation of model-based projections of India's hydrological cycle with geoengineering.

Teaching

Spring 2024 **GEO 202: Ocean, Atmosphere and Climate, Princeton University**
Teaching Assistant: Led precept sections, in-charge of grading and supporting coursework.

Skills & Professional Development

PROGRAMMING: Python (Advanced), Linux (Advanced), Fortran (Basic)

CLIMATE MODELLING: Modifying, running and/or analyzing ocean biogeochemical models (e.g., MOM6-COBALTv2 and MOM6-COBALTv3) and Earth System Models (ESMs, e.g., Max Planck Institute's Grand Ensemble (MPI-GE))

CLIMATE DATA: World Ocean Database, SOCOM flux products, BGC Argo, GLODAP

LANGUAGES: Hindi and English (Native), German (Limited proficiency - B2 Level)

PEER REVIEWER: *Geophysical Research Letters*; *Geoscientific Model Development*; *Journal of Coastal Conservation*; *Journal of Geophysical Research: Oceans*; *Nature Communications*

SERVICE AND OUTREACH

- 2026 **Ocean-Atmosphere Interaction Committee Mentorship Program**, Mentored graduate student at Scripps
- 2025-2026 **Princeton Women in Geosciences Mentorship Program**, Mentored first-year graduate student at Princeton
- Apr 2025 **Princeton Science Outreach 3rd Annual Spring Into Science Event**, Volunteer [Blog post]
- 2024-2025 **Princeton Geosciences Graduate Student Seminars**, Organizer
- Aug 2019 **Initiative für Wissensaustausch, Empowerment und Kultur in Ukraine**, Volunteer

SUMMER SCHOOLS

- Nov 2025 **GOOD-OARS International Summer School**, Universiti Sains Malaysia, Malaysia.
- Aug 2020 **EaSyMS Earth System Modeling School**, MPI-Meteorology & German Climate Computing Centre. Germany.
- July 2020 **Climate-KIC Journey**, European Institute of Innovation & Technology. Estonia, Sweden, and Hungary.

INVITED TALKS

6. **Schmidt Sciences OBVI Early Career Researcher Workshop**. Keynote Speaker, InMOS Project. Boston, September 2026.
5. **European Geosciences Union Ocean Sciences Campfire Series**. Evaluating the impact of bubble-mediated air-sea gas exchange on carbon dioxide and oxygen in the ocean. Online, July 2026.
4. **High Meadows Environmental Institute Discovery Day**. Evaluating the impact of bubble-mediated air-sea gas exchange. Princeton, April 2026.
3. **Schmidt Sciences Integration Across Models and Observations (InMOS) Fortnightly Meeting**. Influence of wind-wave-bubble variability on air-sea CO₂ exchange: Insights from a high-frequency observation-based product. Online, January 2026.
2. **Schmidt Sciences InMOS Fortnightly Meeting**. Influence of wind-wave-bubble variability on air-sea gas exchange. Online, March 2025.
1. **Stockholm Environment Institute**. Women, Water, Climate: Tackling the Challenges Webinar. November 2020. [Link]

Awards

- 2025 **Best Presentation by Student Vote & Best Group Report**, GOOD-OARS International Summer School [News]
- 2024-2026 **Fellow for Climate & Environmental Sciences and Engineering**, High Meadows Environmental Institute
- 2021-2026 **Centennial Fellow in the Natural Sciences and Engineering**, Princeton University
- 2020-2021 **International Student Merit Scholar**, The German Academic Exchange Service (DAAD)
- 2017 **Telenor Youth Forum Winner**, Telenor [Link]
- 2016 **Ministry of Environment Excellence Award Recipient**, Republic of Korea
- 2015-2017 **Cargill Global Scholar**, Cargill & Institute of International Education (IIE)